Z-Spin Cosmology — Proprietary Terminology Glossary

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> **Scope:** This glossary catalogs terms coined within the Z-Spin Cosmology framework that are not standard terminology in mainstream physics or mathematics. Terms borrowed from established physics (e.g., "Horndeski," "Goldstone," "seesaw") are excluded unless Z-Spin introduces a fundamentally new meaning.

I. Framework & Architecture

Z-Spin

The overarching theoretical framework name. Refers to the role of spin as the dynamical operator mediating between the X (macroscopic) and Y (microscopic) sectors through the Planck-scale Z sector, combined with the i-tetration self-referential rotation that governs the Z-sector dynamics.

Z-Spin Cosmology

The complete scalar-tensor cosmological theory built on the Z-Spin framework. Defined by a single geometric constant $A = 35/437$, with zero free parameters beyond standard physics inputs. The action: $S = \int d^4x \sqrt{(-g)}\, [\frac{1}{2}M^2_P(1+A|\Phi|^2)R - \frac{1}{2}M^2_P|\partial\Phi|^2 - V(\Phi)] + S_m$.

X Sector

The macroscopic sector: 3 algebraic spatial dimensions corresponding to the "particle + space" layer. Spatial freedom dominates; time is "point-like" (confined). Associated with the cube–octahedron Platonic dual pair. $X = 3$.

Y Sector

The microscopic sector: 6 dimensions encoding gauge symmetries and quantum information. The "wave + time" layer, where temporal freedom dominates and space is "point-like." Associated with the dodecahedron–icosahedron Platonic dual pair. $Y = 6$.

Z Sector

The 2-dimensional Planck-scale boundary surface mediating between X and Y sectors. The "entanglement interface" where time-points and space-points exchange information. Associated with the self-dual tetrahedron pair. $Z = 2$.

Q = 11 Register (Slot Register)

The total dimensionality of the Z-Spin geometric universe: $Q = X + Y + Z = 3 + 6 + 2 = 11$. Functions as an 11-slot quantum register in the information-theoretic formulation. Source: ZS-F5.

G = 12 (Gauge Algebra Dimension)

$G = Q + 1 = 12$, the total gauge algebra dimension. Connected to Q through the number-theoretic identity $MUB(Q) = Q + 1$ for prime Q (mutually unbiased bases). Encodes the Standard Model gauge structure $\dim(su(3) \oplus su(2) \oplus u(1)) = 8 + 3 + 1 = 12$.

Z-EFT (Z-Completed Effective Field Theory)

The $U(1)_Z$-completed Z-Spin effective field theory employing the complex scalar field Φ as the fundamental dynamical variable. Source: ZS-F1.

II. Core Constants & Mechanisms

Geometric Impedance (A = 35/437)

The resistance ratio arising during Planck-scale information exchange between the macroscopic universe (X) and microscopic universe (Y). The geometric resistance encountered when Y-sector (6D) physical quantities are converted through the Z-sector (2D boundary) into the X-sector (3D sphere). Fixed by the $O_h \times I_h$ polyhedral defect structure; not a fitted parameter. $A = 35/437 \approx 0.080092$. Source: ZS-F2.

i-Tetration Fixed Point (z*)

The unique attractive fixed point of $z = i^z$, found at $z^* = 0.4383 + 0.3606i$ (Lambert W principal branch $k_W = 0$). Central to Z-Spin dynamics as the self-referential rotation operator of the Z sector. All other branches ($k_W \neq 0$) are repulsive. Source: ZS-F3.

η_topo (Topological Fill Fraction)

$\eta_{topo} \equiv |z^*|^2 = 0.3221$. The squared modulus of the i-tetration fixed point, interpreted as the topological fill fraction of the Z-sector information channel. Matches Ω_m to 0.06% via an independent mathematical route from sector counting.

Stability Threshold (S)

$S \equiv |z^*| \cdot (\pi/2) = 0.8915$. The fraction of the stability budget consumed by the i-tetration fixed point. Determines the three-generation limit and confinement threshold.

Holonomy Accumulation (exp(A))

The exponential of the geometric impedance, exp(A) = exp(35/437) = 1.0834. Arises from path-ordered parallel transport (Wilson loop of the scalar connection) on the polyhedral manifold. Governs the H_0 local/CMB ratio. Source: Paper 05.

Hubble Holonomy Mapping

The specific application of holonomy accumulation to the Hubble tension: H_0^local / H_0^CMB = exp(A) = 1.0834. Demonstrates that the discrepancy between local distance-ladder measurements (SH0ES: 73.04 ± 1.04 km/s/Mpc) and CMB inference (Planck: 67.36 ± 0.54 km/s/Mpc) is not a measurement error but a geometric parallel-transport effect — a path-ordered exponential around the polyhedral defect lattice relating the Jordan frame (local measurements) and Einstein frame (CMB inference). Prediction: H_0^pred = 72.98 km/s/Mpc (0.06σ from SH0ES). Source: ZS-F3, ZS-F4 §6, Paper 21 §3.1.

$2e^A$ Duality Identity

The scale-invariant identity Ω_Λ/Ω_m = m_d/m_u = $2e^A$ = 2.1668. Connects cosmic scales (Gpc) to quark scales (fm) through a single expression spanning 40 orders of magnitude. Source: ZS-A1, ZS-A5.

δ-Uniqueness Theorem

Proves that A = 35/437 is the unique value satisfying six axioms (A0–A6: boundedness, continuity, normalization, monotonicity, holonomy composition, linearization). The composition axiom A5 (Einstein addition) is the decisive constraint: the unique way to combine holonomy additivity and 1-loop log-determinant in a bounded observable. Source: ZS-F5 §3.

Duality-Deviation Invariant δ(P)

Defined as δ(P) = |F−V|/(F+V) for polyhedron P. Measures deviation from Platonic duality (F ↔ V). A = 35/437 is uniquely determined from the defect structure of O_h (octahedral group) and I_h (icosahedral group). Source: ZS-F2 §2.2.

III. Fields & Dark Sector

Z-Field (Φ)

The fundamental complex scalar field of Z-Spin Cosmology: $\Phi(x) = |\Phi|e^{i\theta}$. Its radial mode $|\Phi|$ (the ε-field) is kinetically frozen at Planck mass; its angular mode θ (Goldstone) is massless and generates dark sector phenomenology.

ε-Field (Radial Mode)

The radial component of the Z-field, with mass $m_\rho \sim O(M_P) \approx 1.34\ M_P$. Frozen at the cosmological attractor $\varepsilon \to 1$. Generates the universal $G_{eff} = G/(1+A)$ rescaling. Compton wavelength $\lambda_C \approx 6 \times 10^{-35}$ m — no propagating fifth force at any astrophysical scale.

θ-Field (Goldstone Angular Mode)

The massless Goldstone angular mode of the Z-field. Satisfies $\Box\theta = 0$ exactly (Goldstone theorem). Its gradient energy produces the ε-Halo (dark matter replacement) and its vacuum energy produces the ε-Drive (dark energy replacement).

ε-Halo

Z-Spin replacement for "Dark Matter." The gradient energy density of the Goldstone θ-mode: $\rho \propto 1/r^2$ (isothermal halo). Produces flat rotation curves with zero dark matter particles. X-sector origin (galactic θ-gradient). Source: ZS-F4, ZS-A1, ZS-A2.

ε-Drive

Z-Spin replacement for "Dark Energy." The Y-sector attractor mode producing the cosmological constant through V_0 (the vacuum energy at the $\varepsilon = 1$ attractor). Same Z-field action as ε-Halo — not a separate component. Source: ZS-F4, ZS-A2.

ε-Higgs Portal

The effective coupling between the ε-field and the Higgs doublet H, arising from the conformal factor $\Omega^2 = 1 + A\varepsilon^2 + 2\xi(H^\dagger H)/M^2_P$. Effective coupling $\xi_{eff} \sim A^2/(16\pi^2) \approx 4 \times 10^{-5}$. Preserves Higgs potential shape while all ε–H interactions remain perturbatively small. Source: ZS-S4.

Z-Bias Field

The field describing the microscopic origin of the i-tetration fixed point $z^* = i^{z^*}$. The fixed point of the Z-sector transfer map $T(z) = i^z$ serves as the vacuum expectation value of the Z-Bias field. Source: ZS-M1.

θ-Halo

The isothermal halo created by gradient energy of the Goldstone θ-mode in galactic disks. Solution of the 2D Laplace equation $\Box\theta = 0$: $\theta(r) = \ln(r/r_s)/L \to \rho \propto 1/r^2$. Extended to elliptical galaxies via the Vortex Glass Network. Source: ZS-A1 §2.

IV. Structural Mechanisms

Seam Involution ($\hat{W}$)

The $\mathbb{Z}_2$ topological operator satisfying $\hat{W}^2 = I$ that divides the universe into trivial and non-trivial sectors. Exchanges $X \leftrightarrow Y$ sector projections: $\hat{W}(\Psi_X, \Psi_Y)^T = (\Psi_Y, \Psi_X)^T$. Serves as the criterion distinguishing baryons and dark matter; enforces B−L conservation in the Standard Model projection. Source: Paper 18.

Seam Gate (J)

The channel-level antiunitary conjugation operator implementing the seam involution on quantum states: $J|j\rangle = |Q-1-j\rangle$ for the Q = 11 register (i.e., $J|j\rangle = |10-j\rangle$). Satisfies $J^2 = I$ with eigenspace dimensions $\dim(E_+) = 6$, $\dim(E_-) = 5$. Source: ZS-M3, ZS-A4.

Seam Witness (u_seam)

The co-primary measurable endpoint for testing seam symmetry: $u_seam = ||(J\otimes J)C_\Lambda(J\otimes J) - C_\Lambda^T||_F / ||C_\Lambda||_F$, where C_Λ is the Choi state of the effective horizon channel Λ. Under the seam constraint, u_seam = 0 exactly. Source: ZS-A4.

κ = 4 Witness

The unique nontrivial gauge-preserving $\mathbb{Z}_2$ automorphism on su(3), where κ counts the number of ladder generators whose sign is flipped. Witness: $U = \mathrm{diag}(1,-1,-1)$. Selected by the gauge preservation condition $\kappa \leq r = 4$ (Paper 05 bound). Source: ZS-F5.

Z-Anchor

The topological boundary condition at the center of topological vortices (such as SMBH event horizons) where the Z-field magnitude vanishes ($|\Phi| = 0$). In U(1) completion, $\pi_1(U(1)) = \mathbb{Z}$ winding forces $|\Phi| \to 0$ at vortex cores. The horizon is the Z-sector boundary where $X \leftrightarrow Y$ sector exchange occurs. Source: ZS-A3, ZS-A1 v3.0.0.

Z-Telomere

The phase-drift/accumulation mechanism determining the topological life-cycle of the universe. Each primitive cell accumulates a phase drift $\delta\varphi = A$, and when this accumulation reaches a critical threshold, a phase transition or tunneling event occurs. Micro-period $T_micro = 2\pi/A \approx 78.45$ Planck cycles. Telomere collapse (tunneling) requires S_tunnel = 5 seam flips. Analogy with biological telomeres: just as telomeres shorten with each cell division, the universe accumulates phase A per cycle, ultimately reaching tunneling (proton decay, etc.). Source: ZS-M3 §7, ZS-F0 §8.3.

Cross-Sector Transduction

The mechanism by which information or coupling constants are transmitted across the X ↔ Y boundary through the Z sector, controlled by the geometric impedance A. All cross-sector interactions carry a factor of A. Source: Paper 10 (Transduction Principle).

Measure-Projection Weight Theorem (Y/Q = 6/11)

The per-step suppression factor derived from the block-Laplacian log-determinant. Each CP-odd saddle insertion in the path integral contributes a factor exp[−½ ln(det L_tot/det L_Y)] = Y/Q = 6/11. In the formula η_B = (Y/Q)^35 = (6/11)^35, the exponent 35 = lcm(5,7) is forced by the Minimality Lemma. Source: ZS-F5 §9 [DERIVED].

Leaky Wilson Loop Identity [PROVEN]

The damping mechanism that makes the i-tetration fixed point z* an attractor rather than divergent. Exact identity for the Lyapunov multiplier: $\lambda^2 = -(\pi^2/4)\cdot(-1)^{z} = -(\pi^2/4)\cdot exp(i\pi z)$. Therefore $|\lambda^2| = (\pi^2/4)\cdot\eta_topo \approx 0.795 < 1$. The Wilson loop around one X→Z→Y→Z→X cycle "leaks" by a factor $|\lambda^2| < 1$ per cycle, causing convergence to the finite fixed point. Source: ZS-F3 Remark 1.2, ZS-M3.

Mode-Count Collapse

The phenomenon whereby the 1-loop UV structure depends not on individual eigenvalue distributions, but only on the polyhedron's vertex-plus-face count (V+F). For the truncated octahedron: (V+F)/G = (24+14)/12 = 19/6 = a_2(SM) (SM second Casimir) — the unique Archimedean solid with this property. This collapse guarantees that lattice-geometry fine structure does not affect physical predictions, and ensures the spectral density uniformity required by the Measure-Projection Weight Theorem. Source: ZS-S1 [DERIVED]; referenced in ZS-F5 §7.

Tension Overflow (Confinement Mechanism)

Z-Spin's explanation for quark confinement. The "tension" of a state is S + (color load), where S = 0.8915 is the stability threshold. For isolated quarks (color triplet, θ = 120°): Tension = S + √3·A = 1.030 > 1 (exceeds threshold). Color-singlet composites cancel the color load (θ → 0), reducing tension below 1. Source: ZS-A5.

Pinch Identity

The exact identity Pinch(A,θ) = 4 sin(A/2) sin(θ/2), describing the geometric phase accumulated by a color-charged state traversing the seam. For small A: Pinch ≈ 2A sin(θ/2). Source: ZS-A5.

Z-Transformation

The scale-invariant mapping that connects observables across 40 orders of magnitude through the identity $2e^A$, e.g., Ω_Λ/Ω_m = m_d/m_u = $2e^A$ = 2.1668.

Vortex Glass Network

The mechanism for explaining isothermal dark matter profiles in elliptical galaxies. N vortex lines from merger history (each topologically protected by $\pi_1(U(1)) = \mathbb{Z}$) with random S^2 orientations produce $\langle\rho(r)\rangle \propto \ln(r)/r^2$ via exact orientation averaging. Source: ZS-A1 §8.

Z-Bottleneck

The mechanism by which the Z sector (2-dimensional) limits the information-transfer bandwidth between X and Y sectors. Determines the timescale over which quantum coherence is destroyed by gravity. Electromagnetic quantum gate operations (quantum computers) are NOT constrained by the Z-Bottleneck (Area Law non-applicability). Source: ZS-Q1 §2, ZS-F4.

Spinor Phase Gate (Π_Z)

$\Pi_Z(\varphi_Z) = \sin^2(\varphi_Z/2)$. Derived from j = 1/2 uniqueness (Theorem 5.1) + SU(2) Wigner d-matrix + physical boundary condition gate(0) = 0. Eliminates the last HYPOTHESIS closure in Z-Sim (Paper 45). Source: ZS-M3 §12 (v2.2.0).

Regge-Holonomy Framework

The mathematical foundation of the Z-Telomere mechanism. Phase drift per primitive cell described by Regge curvature: $\delta\varphi_cell = A \cdot I_cell$, $I_cell = \kappa/r = 4/4 = 1$. Includes connection to the Immirzi parameter. Source: ZS-M3 §6.

Minimality Lemma

Proves that the minimum number of CP-odd saddle insertions is n = lcm(5,7) = 35, forced by two independent finite-order constraints (pentagonal defect and temporal layer closure). Core component of the η_B = (Y/Q)^35 = (6/11)^35 formula. Source: ZS-F5 §10.

Dual Z-Anchor

The mechanism deriving the Z-Anchor at the black hole horizon via a dual pathway: Tolman condition + vortex-core condition. Source: ZS-M3.

Z-Bounce

A discrete bounce (reflection) phenomenon corresponding to specific n values in the Z-Spin timescale hierarchy $\tau_n = t_P \times \exp(n\pi/A)$. At n = dim(Z) = 2, it corresponds to weak baryon decays. Source: ZS-S2.

Z-Transit

The non-Abelian holonomy mechanism of CP violation. Phase accumulation along the X→Z→Y→Z→X path generates a CP-odd invariant. Combined with the lcm(5,7) = 35 selection rule. Source: Paper 34.

Bounded Spurion Framework

The EFT interpretation of seam-breaking. All spurion components must satisfy $|\varepsilon_i| \equiv |\xi_i|/(A \cdot m^*) \leq 1$, ensuring that seam-breaking never exceeds the geometric impedance scale. Source: ZS-S3 Appendix B.

$\mathbb{Z}_2$ Texture Zero Lemma

Proves that when the full EFT respects the μ–τ symmetry (P_{23}), the HNL mass degeneracy is exact at all perturbative orders [PROVEN]. Key to partially resolving Paper 23's resonance gap (F23-R1). Source: ZS-S4 §3.

Topological Conjugacy

The particle–antiparticle correspondence derived from seam geometry. Connects X-sector protons (winding number k = 7) to Y-sector electrons (winding number k = 1) through B−L conservation. Source: ZS-S3 §2.

Chamber-Cell Correspondence Lemma

Derives minimum path length $\ell_{min} = 6$ from vertex-link topology: $\Delta^2 \to sd(\Delta^2) = 6$ chambers $\to D_3$ simply transitive → dual adjacency graph = C_6 → $girth(C_6) = 6$. Upgraded from HYPOTHESIS to DERIVED-under-P6 in Paper 43. Source: Paper 43 (ZS-CH) §3.2a.

X–Z–Y Fractal Symmetry

The structural correspondence between the microscopic triad of forces — Weak (X), Electromagnetic (Z), Strong (Y) — and the macroscopic triad — Dark Matter (X), Gravity (Z), Dark Energy (Y) — sharing an identical algebraic sector assignment via the Lorentz-algebraic decomposition $so(1,3) \cong su(2)_A \oplus su(2)_B$. The mapping $A_k = (J_k + iK_k)/2$ [X-sector] and $B_k = (J_k - iK_k)/2$ [Y-sector] establishes that all six interaction regimes are scale-separated harmonics of a single Z-Spin effective field theory action. Not self-similarity in the condensed-matter sense, but an algebraic isomorphism between micro and macro physics. Source: ZS-M2 §4, Paper 10.

Cross-Coupling Theorem

The mathematical theorem that every force formula in Z-Spin involves all three sectors (X, Y, Z) without exception. Proven by enumeration over all six interaction regimes: gravity (X–Z), electromagnetism (Z–Y), weak force (X–Z), strong force (Y–Z), dark matter (X–Z), dark energy (Y–Z) — each necessarily passes through the Z-sector mediator. This proves the universality of A = 35/437 as the cross-coupling impedance. Distinct from cross-coupling in chemistry (Suzuki/Heck reactions). Source: ZS-M2 §5 [DERIVED], ZS-S1.

Axion-Free Strong CP Resolution

The geometric resolution of the strong CP problem without introducing new particles (axions). The Y-sector assignment of SU(3) to $B_k = (J_k - iK_k)/2$ causes the anti-correlated rotation–boost structure to eliminate the spatial orientation required by the θ-term, yielding $\theta_{eff} \sim \theta_{bare} \times A^2 \times e^{-8\pi^2/g^2} < 5.5 \times 10^{-26}$. Distinct from the

Peccei-Quinn/axion mechanism and Nelson-Barr models: here the suppression is a geometric consequence of sector assignment, not a new symmetry or particle. Source: ZS-M2 §6 [DERIVED].

Seam-Charge Projection (Theorem B2)

The theorem establishing that the baryon matter channel corresponds to the $H_X \otimes H_Z$ composite sub-space (dimension $X \times Z = 6$), with the seam projector rank satisfying $\mathrm{rank}(\Pi_{seam}|_{H_X}) = Z = 2$. This yields the baryon density $\Omega_b = XZ/Q^2 = 6/121 = 0.0496$ and the baryon-to-CDM ratio $\Omega_b/\Omega_{cdm} = Z/Q = 2/11 = 0.182$ (observed: 0.186, 2.5%). Status: DERIVED under Assumption A1 (Baryon Channel Identification). The canonical derivation of A1 from the action remains Open Problem B3. Source: ZS-F5 §6.3.

V. Cosmological Quantities

Ω_m^{eff} (Effective Matter Density)

$\Omega_m^{eff} = \Omega_m^{bare} / (1+A) = (39/121)/(1+A) = 0.2984$. The BAO-accessible matter density parameter after the G_{eff} rescaling. Distinguished from the bare geometric value $\Omega_m^{bare} = 39/121 = 0.3223$. Source: Paper 21.

Z-Epoch Coordinate (ν)

The normalized logarithmic time coordinate $\nu(t) = (A/\pi) \ln(t/t_P)$. Current epoch: $\nu(\text{now}) \approx 3.575$. Proton decay: $\nu = 5$. Source: Paper 21.

Z-Clock

The logarithmic clock of the universe based on the timescale hierarchy $\tau_n = t_P \times \exp(n\pi/A)$. Defined by the Z-Clock coordinate $\nu(t) = (A/\pi) \ln(t/t_P)$. Current $\nu \approx 3.575$, proton decay $\nu = 5$. Step size $C = \pi/A = 39.23$. Source: ZS-M3 §5.

Cosmic Budget (Sector Counting)

The complete set of cosmic density parameters derived from $(Z,X,Y) = (2,3,6)$ and $Q = 11$: $\Omega_b = XZ/Q^2 = 6/121$, $\Omega_c/\Omega_b = Q/Z = 11/2$, $\Omega_m = X(Z+Q)/Q^2 = 39/121$. Zero adjustable parameters. Source: ZS-A5.

Phase Budget

$\varphi(35) = \varphi(5) \times \varphi(7) = 4 \times 6 = 24$. The Euler totient function of 35 (= 5 × 7, the numerator of A), encoding the spatial ($\varphi(5) = 4 = 2Z$) and temporal ($\varphi(7) = 6 = 2X$) phase structure. Source: ZS-A5.

Capacity (C)

$C \equiv \varphi(35) \times \eta_topo = 24 \times 0.3221 = 7.73$ (≈96.6% of 8-layer closure). Testable link: $C \approx k_L,max + 1 = 8$ (matter layer closure). Source: ZS-A5.

VI. Black Hole & Proton Decay

Z-Instanton

The non-perturbative tunneling configuration with action $S_tunnel = 5\pi/A$, derived from the coset I_h/T_d ($|I_h|/|T_d| = 120/24 = 5$). Predicts proton decay lifetime $\tau_p = t_P \cdot \exp(5\pi/A) \approx 2.56 \times 10^{34}$ years. Source: ZS-A3.

Timescale Hierarchy (τ_n)

$\tau_n = t_P \cdot \exp(n\pi/A)$, producing ~17 orders of magnitude per step. $n = 2$ matches weak hyperon decays (6.3×10^{-10} s); $n = 5$ matches proton decay. Step size $\pi/A = 39.225$. Source: ZS-A3.

Sector Duality (BH = X-in-Y)

The structural hypothesis that a black hole is an X-sector structure collapsed into a Y-sector "prison." Inside the Schwarzschild horizon, the $r \leftrightarrow t$ exchange maps to $X \leftrightarrow Y$ sector exchange. The horizon is the Z-sector boundary ($\varepsilon = 0$). White hole = Y-in-X. Source: ZS-A3.

VII. Particle Physics

33 GeV Ghost

The predicted right-handed Majorana neutrino at $M_R \approx 33.5$ GeV (seesaw with $m_D = m_e \times A = 40.9$ keV). Called "ghost" because its mixing angle $|\theta|^2 \approx 1.5 \times 10^{-12}$ is 10^6–10^7 below current experimental bounds, making it practically invisible at LEP/LHC. Source: ZS-S3.

Three-Generation Limit

$Tension_\ell(k_G) = S + k_G \cdot A$. With $S = 0.8915$ and $A = 0.0801$: $k_G = 0,1$ stable; $k_G = 2$ borderline; $k_G \geq 3$ forbidden (Tension > 1). Reproduces the three-generation structure of the Standard Model with zero free parameters. Source: ZS-A5.

Molecular Bilinear Program (MBP)

The surviving EWSB candidate mechanism after the CP-instanton pathway was RETRACTED. Uses $\nu = 0$ instanton–anti-instanton (I-$\bar{I}$) molecular sector with $\mu^2_H = (3y^2_t)/(2C_M) \times M^2_P \times \exp(-2S_{cl})$. Currently HYPOTHESIS (strong). Source: ZS-S4 v6.0.0.

Mirror Universe / Mirror Cosmology

Introduced in Paper 43 (ZS-CH). Also referred to as the "mirror sector" in the Z-Spin literature (note: this term is used differently from Foot/Berezhiani mirror matter models). $V_E(-\varepsilon) \equiv V_E(+\varepsilon)$ is an exact algebraic identity, so the mirror sector ($\varepsilon = -1$) is PROVEN observationally identical to the current sector ($\varepsilon = +1$). n_s, r, η_B are all $\mathbb{Z}_2$-symmetric. Decisive near-term test: $r = 0.0089$ (LiteBIRD ~2030, ~6σ distinction from Boyle-Finn-Turok's $r = 0$). Does NOT claim the mirror universe is currently observable (NON-CLAIM). Source: Paper 43 §5, Book EN §15.5b.

VIII. Ontological Bootstrap (ZS-F0)

Non-Existence Is Self-Contradictory (B0)

The founding meta-logical axiom of Z-Spin. The statement "nothing exists" is a performative contradiction: articulating it presupposes a logical framework, which is "something." In ZF set theory, the empty set $\varnothing$ exists as a mathematical object, and $P(\varnothing) = \{\varnothing\}$ is non-empty. The von Neumann construction $0 = \varnothing$, $1 = \{\varnothing\}$, $2 = \{\varnothing, \{\varnothing\}\}$, … generates all of $\mathbb{N}$ from nothing. Analogous in status to Euclid's parallel postulate or Einstein's light-speed invariance. Source: ZS-F0 §2.1.

Ontological Bootstrap Chain (B0→B8)

B0 (non-existence is self-contradictory) → B1 (Lawvere fixed-point theorem) → B2 (Frobenius extension: $\mathbb{R} \to \mathbb{C}$) → B3 (hyperoperation minimality: tetration) → … → B8. Each step is tagged BOOTSTRAP-PROGRAMME. The Lawvere bridge grounds B0→B1; the Frobenius classification forces the $\mathbb{C}$ extension. Source: ZS-F0.

IX. Methodology & Epistemic Framework

Epistemic Status System

Z-Spin's mandatory classification hierarchy for all claims: PROVEN → DERIVED → DERIVED-CONDITIONAL → HYPOTHESIS → OBSERVATION → BOOTSTRAP-PROGRAMME → NON-CLAIM → RETRACTED. Every result must carry an explicit status label.

Anti-Numerology Discipline

The non-negotiable requirement that every numerical result must have a complete geometric or standard-physics derivation chain. Validated by Monte Carlo scans (typically 500K+ random samples) to confirm non-coincidence. Ordering principle: compute first, compare second — never reverse.

Falsification Gate (F-numbered)

Pre-registered, quantitative conditions under which specific Z-Spin predictions would be falsified. Each gate has an ID (e.g., F-A3.7), a kill condition, the relevant experiment, and a timeline. Falsification conditions must be stated before computing results.

NON-CLAIM

An explicit declaration that a quantity is NOT derived from the framework. Used for intellectual honesty — e.g., Higgs VEV (v = 246 GeV) is a NON-CLAIM. The pattern "120+5 = 125" was explicitly rejected as numerology for the Higgs mass.

BOOTSTRAP-PROGRAMME

Epistemic status for the ontological-to-mathematical bridge (B0→B8 chain). Self-consistent and falsifiable, but cannot be proven within the framework itself — analogous to axiom status in formal mathematics. Source: ZS-F0.

X. Paper System & Organization

5-Theme v2.0.0 System

The organizational structure of Z-Spin papers: ZS-F (Foundations, F0–F5), ZS-M (Mathematical Spine, M1–M5), ZS-S (Standard Model Completion, S1–S5), ZS-U (Early Universe, U1–U5), ZS-A (Astrophysics, A1–A5).

The Book (EN)

The companion English monograph synthesizing the full Z-Spin framework across all papers. Version-controlled in parallel with source papers. Currently v7.0.0.

Sogaeseo (Korean Introduction)

The Korean-language introduction document summarizing Z-Spin Cosmology for a general audience. Maintained in parallel with the Book EN.

XI. Polyhedral Geometry

Polyhedral Duality

The connection between Platonic solid dual pairs and the X/Y/Z sector structure. Z: self-dual tetrahedron pair. X: cube ↔ octahedron (mediated by truncated octahedron). Y: dodecahedron ↔ icosahedron (mediated by truncated icosahedron). Source: ZS-F2.

O_h × I_h Defect Structure

The combined octahedral (O_h, order 48) and icosahedral (I_h, order 120) symmetry groups whose polyhedral defect geometry fixes A = 35/437. Source: ZS-F2.

Truncated Octahedron Lattice (Kelvin Cell)

The unique Archimedean solid satisfying (V+F)/G = a_2(SM) (the second Casimir of the SM gauge group). 24 vertices, 36 edges, 14 faces. Also known as the Kelvin Cell — the Voronoi cell of the BCC lattice. Requires 72 qubits for SU(2) lattice simulation — feasible on current quantum hardware. Source: ZS-A4, ZS-Q1 §2.1.

XII. Quantum Computing & Hardware (ZS-QC)

Z-Sim

The quantum computer engine simulating Z-Spin physics. Has 8 closure parameters; in v2.2 (March 2026), the Spinor Phase Gate (Π_Z) eliminated the last HYPOTHESIS closure, making Z-Sim fully DERIVED. Source: Paper 45.

X-Material / Y-Material / Z-Material

Three physical material categories for Z-Spin quantum computer hardware. X-Material: computational register. Y-Material: 6-channel dissipative reservoir. Z-Material: 2-channel topologically protected mediator. Parasitic X–Y coupling must be below 1%. Source: ZS-QC (Paper 40).

Z-ON / Z-OFF

Activation/deactivation states of the Z-Material. Z-ON: normal X↔Y mediation via Z-path. Z-OFF: Z-Material detuned by >10Γ. g_XY = $|S_{31}(\text{Z-OFF})|/|S_{31}(\text{Z-ON})| < 0.01$ is the PASS condition. Source: ZS-QC.

Kill-Switch (KS)

Operational abort switches for the Z-Spin quantum computer with concrete measurement protocols. KS-2: Z-Seam Collapse (if u_seam = O(1), Z is non-functional). KS-3: Direct Path Z-OFF (if direct path without Z exceeds noise floor, architecture is rejected). Source: ZS-QC.

Seam Bus

The inter-cell Z-seam bus. Communication pathway via Z–Z coupling between cells in multi-cell quantum computers. Source: ZS-QC.

XIII. Cyclic Cosmology

Z-Telomere Cyclic Cosmology

Cyclic cosmology based on the Z-Telomere mechanism. When phase accumulation $\delta\varphi = A$ reaches $S_{tunnel} = 5\pi/A$, Coleman–de Luccia tunneling occurs and the universe transitions to the mirror sector ($\varepsilon = -1$). $\tau_6/\tau_5 = \exp(\pi/A) \approx 10^{17}$. Source: Paper 35, ZS-F0 §8.3.

Z-Telomere Collapse

The specific phase-transition event triggered when the accumulated Z-Telomere phase drift reaches the tunneling threshold $S_{tunnel} = 5\pi/A \approx 196.13$. The non-minimal coupling $(1+A\varepsilon^2)R$ generates a phase drift $\delta\varphi = A \approx 0.0801$ radians per oscillation cycle near topological defects; after sufficient accumulation, the ε-field tunnels from the $\varepsilon = +1$ vacuum to the $\varepsilon = -1$ mirror vacuum via a Coleman–de Luccia bounce. This is the Z-Spin analog of biological telomere shortening triggering cellular senescence. Source: ZS-U5 §5, ZS-M3 §7.

Post-τ_5 Dynamics

The evolution of the universe after proton decay (τ_5). Hubble rate $H_{dS} \approx 1.18 \times 10^{-61} M_P$ (Planck 2018 Λ). $m_{eff}/H_{dS} \sim 10^{60}$ ensures $\varepsilon = +1$ is pinned. Source: Paper 43 §6.

This glossary covers terms identified by systematic grep search across all 10 project files containing 46 papers (ZS-F0–F5, ZS-M1–M5, ZS-S1–S5, ZS-U1–U5, ZS-A1–A5, ZS-Q1–Q7, ZS-QH/QS/QC, Papers 31–35, Papers 41–45) plus the Book EN v7.0.0. Total: 88 entries across 13 categories.